

International Cybersecurity and Digital Forensics Academy

SentinelNet AI Defense Fabric

AICS 109 Capstone Report

AI for Advanced Network Defense

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AI Driven Cybersecurity and Digital Forensics Fellowship

Student: ijudigal jarafu | Registration No: 2026/AICDF/16789

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1 Executive Summary

SentinelNet is a controlled-lab network defense prototype that converts flow telemetry into evidence-rich alerts for analyst review. The final implementation trains and reloads a residual multilayer perceptron, an autoencoder, an Isolation Forest, and a gated recurrent unit model; combines their evidence with telemetry, sensor, and asset context; maps candidate threats to ATT&CK; and emits recommendations that cannot execute network changes.

The final run generated 12,000 deterministic synthetic flows and processed the first 1,000 flows through an end-to-end streaming integration demonstration. It produced 384 alerts at the configured medium threshold of 65. This stream overlaps the model-development corpus and is not an independent evaluation set. Held-out model results were: supervised macro F1 0.9987, autoencoder anomaly macro F1 0.9595, Isolation Forest macro F1 0.7942, and sequence GRU macro F1 0.1223. The low GRU result is retained and explained because the synthetic labels do not contain realistic campaign-level temporal dependence.

Recommendation: treat the system as a reproducible engineering demonstration, not a production control. The next validation must use authorized real telemetry, realistic attack sequences, calibrated thresholds, drift monitoring, and analyst-approved response integration.

Outcome	Evidence
Reproducibility	Seed 42, one-command pipeline, version and timing log
Evaluation	Separate test data; training-only preprocessing and threshold selection
Explainability	Every alert exposes six risk components and supporting telemetry
Safety	Dry-run recommendations with mandatory human approval
Verification	38 automated tests and deliverable consistency validation

2 Lab Environment

The final pipeline ran on Ubuntu 22.04 with Linux kernel 6.8.0 on an x86_64 host. The machine exposed four Intel Celeron N2920 cores at 1.86 GHz, 7.6 GiB RAM, and 2.0 GiB swap. Training used CPU-only PyTorch.

Component	Recorded value
Python	3.10.12
pandas	2.3.3
NumPy	2.2.6
scikit-learn	1.7.2
PyTorch	2.14.0+cpu
joblib	1.6.0
pytest	9.1.1

Project structure: scripts/ contains pipeline stages and shared model code; data/ contains deterministic flow telemetry; models/ contains loadable checkpoints and preprocessing metadata; outputs/ contains evaluation and alert evidence; tests/ contains automated checks; and submission/ contains the final report and defense materials.

3 Dataset and Telemetry

The generator produced 12,000 rows with 22 columns and no missing values. Class counts were Benign 7,401, PortScan 1,189, DDoS 992, BruteForce 932, Exfiltration 772, and C2Beacon 714. Timestamps begin at 2026-01-01 and the seed is fixed at 42, so reruns produce the same records.

The schema approximates enriched Zeek or Suricata flow context but does not ingest live sensors. It contains source and destination addressing, ports, protocol, duration, packet and byte volumes, derived rates, five-minute scan and authentication counters, DNS entropy, beacon and burst indicators, Suricata alert counts, TCP state, asset criticality, and a synthetic label.

Ethics and privacy: the dataset is synthetic, avoids collection of personal or unauthorized traffic, and uses private or documentation-range addresses. That reduces privacy risk but also reduces external validity. Real telemetry must be collected under authorization, minimized, access-controlled, and retained according to policy.

4 Feature Engineering

Fourteen numeric features feed all three neural model paths. Inputs with missing or non-finite required features are rejected before splitting. The split occurs before preprocessing and the scaler is fitted on training rows only, preventing test statistics from leaking into model training.

Feature group	Fields	Defensive relevance
Volume and duration	duration, packets, bytes_out, bytes_in	Capture transfer size and connection persistence
Rates and direction	bytes_per_sec, packets_per_sec, outbound_ratio	Separate bursts, floods, and outbound-heavy transfer
Local behavior	unique_dst_ports_5m, failed_conn_5m	Expose scan and repeated-authentication patterns
Application indicators	dns_entropy, beacon_score, burst_score	Represent tunneling, periodicity, and burst behavior
Context	suricata_alert_count, asset_criticality	Add sensor corroboration and business impact

5 Supervised Model

The supervised classifier is a residual MLP with 14 inputs, a hidden width of 96, batch normalization, ReLU activations, dropout 0.15, two residual blocks, and six output logits. Training used eight epochs, batches of 256, seed 42, 9,000 training rows, and a stratified 3,000-row test partition.

Held-out performance was macro F1 0.9987, weighted F1 0.9993, and accuracy 0.9993. False-positive analysis: the confusion matrix contained only two errors, both PortScan flows predicted as BruteForce; this created two BruteForce false-positive predictions and no false-positive predictions for the other attack classes. The high score reflects deliberately distinct synthetic class distributions and must not be interpreted as production performance.

Confusion matrix rows are actual classes and columns are predicted classes.

Actual	Benign	Brute	C2	DDoS	Exfil	Scan
Benign	1850	0	0	0	0	0
Brute	0	233	0	0	0	0
C2	0	0	179	0	0	0
DDoS	0	0	0	248	0	0
Exfil	0	0	0	0	193	0
Scan	0	2	0	0	0	295

6 Anomaly Model

The autoencoder uses a 14-64-24-8 encoder and a symmetric 8-24-64-14 decoder. It was fitted on 4,440 benign training rows. The decision threshold, 0.05419, is the 95th percentile of reconstruction error from 1,111 held-out benign training-validation rows. The test set was not used to choose this threshold.

On the 3,000-row test set, the autoencoder achieved macro F1 0.9595 and weighted F1 0.9613. The Isolation Forest achieved macro F1 0.7942 and weighted F1 0.8112. The autoencoder is the stronger anomaly model here; the Isolation Forest remains useful as a structurally different corroborating detector.

False positives are expected when legitimate behavior differs from the benign training baseline. Thresholds must therefore be recalibrated per environment, and analysts should inspect raw sensor context before response.

7 Sequence Model

The sequence model is a one-layer PyTorch GRU with input dimension 14 and hidden dimension 48. Each training example uses the preceding eight chronologically ordered flows to predict the next flow label. The chronological 75/25 split is applied before window construction, so no sequence crosses the train/test boundary.

After six epochs, the GRU achieved macro F1 0.1223, weighted F1 0.1795, and accuracy 0.1494 across 2992 test sequences. This weak result is expected because labels in the synthetic generator are independently sampled rather than organized into sessions or campaigns. The model and checkpoint are real; the dataset does not provide enough valid temporal signal.

The sequence component remains capped at 10 risk points and contributes in proportion to confidence. Future sequence evaluation should use campaign-aware data with beacon intervals, scan bursts, authentication attempts, and staged exfiltration.

8 Alert Fusion and ATT&CK Mapping

The streaming integration demonstration processed 1,000 flows and emitted 384 alerts at risk score 65 or higher. These rows overlap the model-development corpus, so the alert count demonstrates integration behavior rather than independent generalization. The 100-point formula is supervised 45, anomaly 20, sequence 10, telemetry 10, sensor 5, and asset criticality 10. The stored component values must sum to the displayed score; the validator checks this invariant.

Evidence	Maximum	Purpose
Supervised	45	Known-pattern classification probability
Anomaly	20	Autoencoder and Isolation Forest deviation evidence
Sequence	10	Confidence from preceding-flow context
Telemetry	10	Threat-specific flow indicators
Sensor	5	Suricata alert corroboration
Asset	10	Destination criticality

ID	Threat	Evidence	ATT&CK	Recommended action	Reason
1	DDoS	risk 88; model 0.9999; anomaly 1.00	T1498 Network Denial of Service	Validate telemetry; recommend lab isolation after approval	components 45+20+2+6+5+10
2	DDoS	risk 88; model 0.9999; anomaly 1.00	T1498 Network Denial of Service	Validate telemetry; recommend lab isolation after approval	components 45+20+2+6+5+10
3	DDoS	risk 88; model 0.9996; anomaly 1.00	T1498 Network Denial of Service	Validate telemetry; recommend lab isolation after approval	components 45+20+2+6+5+10

ATT&CK mappings provide investigation context, not proof of adversary technique or attribution. The analyst must validate the original flow, related events, host context, and sensor evidence.

9 AI Response Recommendation

The simulator generated 20 response recommendations from the highest-priority alerts. Every action is labeled dry-run and requires human approval. It cannot alter firewalls, isolate endpoints, disable accounts, or invoke a production SOAR platform.

For RFC1918 source addresses, a high-risk alert may produce `recommend_isolate_lab_host`. This is an analyst recommendation only. Public, documentation-range, or malformed sources are escalated rather than isolated, preventing the prototype from proposing direct action against an address outside the controlled lab boundary.

- Validate the raw flow and surrounding Zeek or Suricata events.
- Confirm the source is an authorized lab asset and assess business impact.
- Document analyst approval before any containment in a separate authorized system.
- Record false positives and feed threshold review rather than silently suppressing alerts.

10 Limitations and Future Work

The model was trained in a controlled lab and needs validation on real authorized telemetry.

The response simulator is not a production SOAR system.

High confidence does not remove the need for analyst review.

Dataset quality and label quality directly affect model reliability.

Additional limitations include class patterns that are easier to separate than real traffic, independently sampled labels that weaken temporal evaluation, uncalibrated probabilities, synthetic sensor fields, no concept-drift monitor, and no adversarial-evasion assessment. Real deployment would also require privacy review, access control, retention limits, model governance, rollback procedures, and measurement against operational alert base rates.

Prioritized future work is to ingest authorized Zeek or Suricata records, define campaign-aware temporal labels, compare against simple baselines, calibrate scores and thresholds, measure precision at realistic prevalence, add drift and data-quality monitoring, and integrate only with approval-gated case management.

Completed Model Comparison

Model	Input features	Best metric	Strength	Weakness
Residual MLP	14 flow features	Macro F1 0.9987	Known synthetic patterns	May overstate real performance
Autoencoder	Benign baseline	Macro F1 0.9595	Strong deviation signal	Threshold is environment-specific
Isolation Forest	Benign baseline	Macro F1 0.7942	Non-neural corroboration	Lower attack recall
GRU Sequence	Eight-flow windows	Macro F1 0.1223	Real temporal pipeline	Data lacks temporal structure

Appendix A Commands

The final profile reproduces the complete evidence set and records each command, exit code, elapsed time, package version, and platform detail in outputs/execution_log.txt.

```
.venv/bin/python scripts/09_run_pipeline.py --profile final
.venv/bin/python -m pytest tests -q
.venv/bin/python scripts/08_validate_deliverables.py
git diff --check
```

Appendix B Outputs

Artifact	Purpose
data/synthetic_flows.csv	Deterministic 12,000-row dataset
models/supervised_ids.pt	Residual MLP checkpoint
models/autoencoder.pt	Autoencoder checkpoint
models/isolation_forest.joblib	Isolation Forest artifact
models/sequence_gru.pt	GRU checkpoint
outputs/*_metrics.json	Model evaluation evidence
outputs/stream_alerts.jsonl	384 fused alert records
outputs/fusion_summary.json	Risk formula and top alerts
outputs/response_plan_lab_only.json	Dry-run response recommendations
outputs/execution_log.txt	Versions, commands, timings, exit codes
submission/defense_notes.md	Oral defense preparation
submission/presentation_outline.md	Eleven-slide presentation structure

Evidence integrity: scripts/08_validate_deliverables.py confirms that required files exist, JSON and JSONL parse correctly, alert counts agree, risk components sum correctly, and every response remains dry-run with human approval.